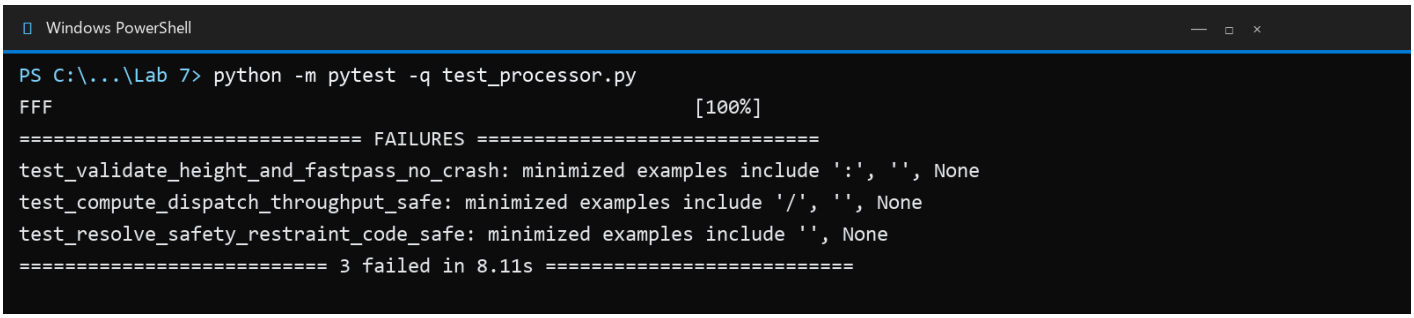
Lab 7: Fuzz Testing with Hypothesis

Student SRN: PES2UG24CS013
Repository: L6_L7_11_amusement_park_ride_access
Objective: Use Hypothesis to find edge-case failures, harden three ride-access functions, and verify the fixes.

Before the fix

The starter crashed on malformed rider specifications, invalid dispatch metrics, and unknown restraint zones. Hypothesis minimized failures to inputs including `:`, `/`, empty strings, and `None`. The captures show the baseline app and failing fuzz run.





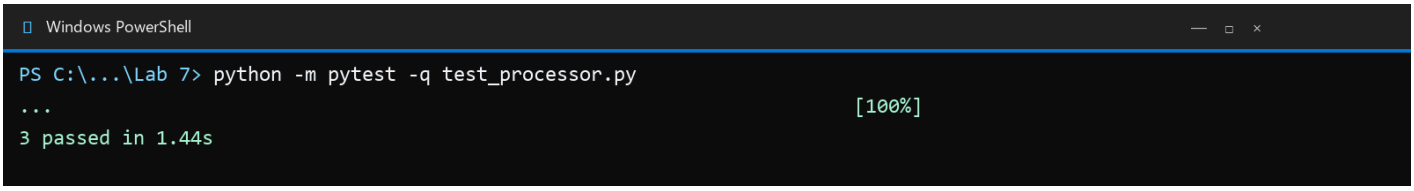
Fix and re-run

Input type and structure are checked before parsing. Invalid rider specifications return a clear message, invalid throughput input returns `None`, and unknown zones return `Unknown ride zone`. Valid input behavior is retained. The final Hypothesis suite exercises arbitrary strings and `None` for all three functions.

Fixed application output



Hypothesis tests after the fix



Reproduce from the Lab 7 folder

```
python processor.py
python -m pytest -q test_processor.py
```